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# Ultrasound Findings of Non-Mass Type Breast Tuberculosis: Beyond Typical Lesions

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## BACKGROUND

Breast tuberculosis can be either primary or secondary, but primary breast tuberculosis is extremely rare. If there is no other focus of infection elsewhere in the body, it is considered primary. The diagnosis of breast tuberculosis has three essential pillars: clinical examination, radiological evaluations, and histopathological sampling. Radiological evaluations are not only critical in diagnosis but are also important in the planning of the treatment and during the follow-up.

## Case Presentation

A 26-year-old unmarried woman presented with a lump in her left breast and left axilla that had been present for six months. The lumps were enlarging and painful. The patient's mother had recently completed treatment for pulmonary tuberculosis the previous year. There were no respiratory complaints or history of pulmonary tuberculosis.

Ultrasound findings revealed a hypoechoic lesion with ill-defined margins and focal distribution, accompanied by lactiferous duct dilatation and a sinus tract; no calcification or architectural distortion

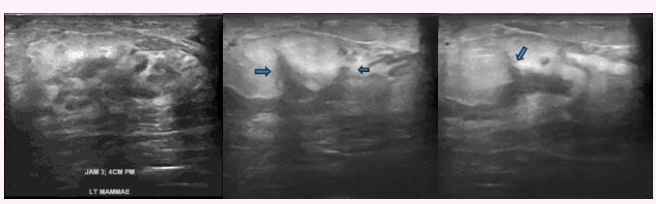



Figure 1. Sonography show hypoechoic lesion with ill-defined margins and focal distribution accompanied by lactiferous duct dilatation and a sinus tract (*arrows*)

Multiple benign node lymphadenopathies with uniform cortex thicker than 3 mm, minor surface lobulation and some lacking a fatty hilum.




Figure 2. Multiple lymphadenopathies with thickening cortex and some lacking a fatty hilum.

Histopathological analysis showed features suggestive of a specific chronic tuberculous inflammatory process with abscess formation.

Microscopic pathological findings reveal clusters and scattered inflammatory cells including lymphocytes, neutrophils, macrophages, and multinucleated giant cells within the connective tissue stroma and surrounding the ducts, lobules, and adipose tissue. Vascular proliferation with dilated, congested lumens is observed, alongside interstitial hemorrhage and necrotic masses in several foci.

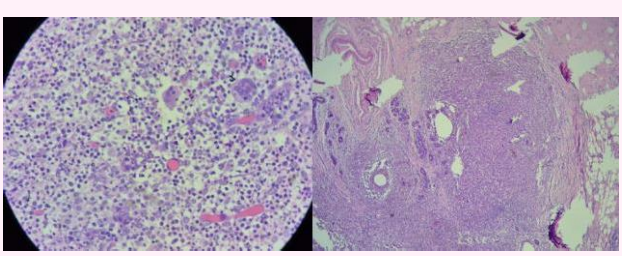


Figure 3. Microscopic pathological findings inflammatory cells including lymphocytes, neutrophils, macrophages, and multinucleated giant cells. Vascular proliferation with dilated, congested lumens is observed, alongside interstitial hemorrhage and necrotic masses in several foci.

## Discussion

Ultrasonography is often the first radiological examination method used in diagnosis.

It is also the best method for evaluating axillary lymph nodes.

Breast tuberculosis lesions can be classified into nodular, disseminated, sinus, or abscess type.

Tuberculosis mastitis and Granulomatosis mastitis have similar clinical symptoms including solitary breast mass, Ulcer, chronic discharging sinus or an abscess. A breast mass is the common presentation in Granulomatosis mastitis.

Patients commonly present with lump, breast pain and overlying skin erythema usually present in unilateral but also present in bilateral

Ultrasound of Granulomatosis mastitis showed hypoechoic tubular or nodular mass, hypoechoic nodular structure, parenchymal distortion without mass lesion, lobulated mass, multiple clustered often contiguous with large hypoechoic mass and enlarged axillary lymhpnode.



Figure 4. Sonograms show large irregular hypoechoic mass with multiple tubular extensions (*arrows*).

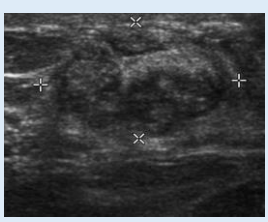


Figure 5. Sonographic findings heterogeneous hypoechoic mass

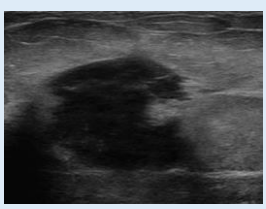


Figure 6. Sonograms of axilla show abnormal lymph node.

## CONCLUSION

Breast tuberculosis is a difficult disease to diagnose, and clinical evaluation, radiological examination, histopathological and microbiological sampling are often necessary to confirm diagnosis.

Radiological examinations are not only important in diagnosing the disease but are also important during the treatment and in the follow-up.

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